

IC3Net - Condensed Paper Report

Paper: Learning when to Communicate at Scale in Multiagent Cooperative and Competitive Tasks

Authors: Amanpreet Singh, Tushar Jain, Sainbayar Sukhbaatar

Venue: ICLR 2019 / arXiv:1812.09755

Source: <https://arxiv.org/pdf/1812.09755>

Scope of this report: What this paper contributes beyond CommNet, what its experiments actually show, its gaps, and the experiment/implementation considerations it creates for our project. It does not set the overall project direction.

1. What the paper mostly talks about

- **Problem.** Always-on continuous communication works well in fully cooperative tasks, but it can be harmful or unnecessary when agents have different incentives. The paper asks whether agents can learn when communication is beneficial.
- **Proposed model.** IC3Net extends continuous hidden-state communication with a learned binary communication gate and trains agents using individualized rewards.
- **Target settings.** The same architecture is evaluated in cooperative, mixed/semi-cooperative, and competitive scenarios, with emphasis on larger teams/maps and partial observability.
- **Two claimed improvements.** (1) individualized rewards improve credit assignment and scalability; (2) gating lets an agent suppress its outgoing communication when sharing information is not profitable.

2. Core IC3Net mechanism

- **Recurrent per-agent controller.** Agents share an LSTM model. The shared parameters make the controller symmetric/permutation tolerant while each agent keeps its own recurrent hidden state.
- **What is communicated.** The communication signal is derived from other agents' hidden states, averaged and passed through a learned linear transformation.
- **When communication occurs.** Each sender produces a binary gate g (communicate / do not communicate). Only hidden states from agents with active gates contribute to the shared communication vector.
- **How the gate is learned.** The binary gate and the environment action policy are trained with REINFORCE; the continuous message path remains differentiable.
- **Reward change.** Unlike CommNet's single global reward view, each agent is optimized with its own reward. This is the paper's main credit-assignment intervention.
- **Timing detail.** The gate used for the next time step is computed from the current hidden state.

Critical reading point: IC3Net changes both the reward/credit-assignment scheme and the communication mechanism. Therefore, better performance versus CommNet cannot automatically be attributed to the gate. The paper itself provides evidence that individualized rewards account for a large part of the gain.

3. Experiments and major findings

Environment / test	What it isolates	Main finding
Predator-prey: cooperative / mixed / competitive	Whether the gate changes with incentives.	Cooperative predators communicate almost always; prey learns silence; competitive predators communicate only while it is profitable.
Predator-prey scaling	Larger grids + more agents; compare IRIC, IC, CommNet, IC3Net.	IC3Net's advantage over CommNet becomes large at 20x20 / 10 agents (52.4 vs 76.5 average steps; lower is better).
Traffic junction	Zero-vision coordination across easy/medium/hard versions.	IC3Net matches CommNet on easy but strongly outperforms it on medium/hard

		(89.3% / 72.4% vs 54.3% / 50.2%).
StarCraft exploration	Larger realistic partially observed search task.	IC3Net substantially outperforms CommNet on exploration, especially at larger map scale.
StarCraft combat	Cooperative combat where always sharing is useful.	IC3Net is roughly comparable to CommNet rather than clearly better; gating is not universally beneficial.
Gate/reward ablation	Always-on communication + individualized reward.	Individual rewards alone help greatly; on one mixed 20x20 predator-prey ablation, always-on communication + individual reward is slightly better than learned gating, showing gate overhead.

4. Conclusions we can safely take from the paper

- “When to communicate” can be learned as a separate policy decision rather than hard-coded.
- Communication policy should depend on incentives and task phase: agents may rationally communicate, remain silent, or stop communicating after information becomes strategically harmful.
- Credit assignment is a major confound in MARL communication. Individualized rewards can improve learning even without a sophisticated communication mechanism.
- The usefulness of gating is conditional. When a task is fully cooperative and communication is almost always useful, forcing the gate open can train faster than learning a gate.
- Gating becomes more meaningful when communication has a downside: competitive leakage, unnecessary communication, or some other cost/constraint.
- Scalability claims should be evaluated jointly with reward design, not just communication architecture.
- A single communication model can work across cooperation regimes, but the reward function determines what “profitable communication” means.

5. Gaps, flaws, and limitations

- **Sender-only broadcast gate.** The model decides whether to speak, but not whom to speak to. If active, the message is still effectively broadcast/aggregated rather than targeted per recipient.
- **No receiver-side choice.** Agents cannot explicitly choose whether to listen. Multi-channel and listen/no-listen decisions are named by the authors as future work.
- **No explicit communication resource model.** There is no bandwidth budget, latency, packet loss, noise, or energy cost. In cooperative tasks, this often drives the optimal gate toward always-on.
- **Reward design may manufacture the gating behavior.** The competitive/mixed reward structure determines whether revealing information is profitable. This is useful, but it is not the same as learning efficient communication under a physical communication budget.
- **Gating and individualized rewards are entangled.** Several gains are attributable to improved credit assignment. The ablation shows the gate can add optimization overhead.
- **Binary gate uses high-variance policy gradient.** The gate is trained with REINFORCE rather than a differentiable relaxation or modern actor-critic/PPO approach.
- **Broadcast averaging still dilutes identity.** The receiver does not get a clean sender-specific message stream; targeted addressing and message routing remain open problems.
- **Combat result is not a universal win.** On StarCraft combat, IC3Net is comparable to CommNet, reinforcing that “learned silence” is task-dependent.
- **Limited robustness/generalization testing.** The paper scales selected environments but does not deeply test unseen maps, heterogeneous roles, bandwidth shifts, message corruption, or latency.

6. Implementations / experiments this paper makes us consider

- **Separate WHAT from WHEN.** Keep the continuous message generator and a separate communication gate so we can test whether timing decisions add value over the same message architecture.

- **Do the reward/communication ablation properly.** At minimum compare: no comm + team/global reward; no comm + individualized/auxiliary credit; always-on comm + same reward; gated comm + same reward. Otherwise we cannot tell whether communication or credit assignment caused the improvement.
- **Include a forced-open gate baseline.** In strongly cooperative tasks, $g=1$ is a necessary control because the paper found that learning the gate can slow training when communication is almost always useful.
- **Log communication rate as a first-class metric.** Track percentage of agents transmitting, messages per episode, and how gating changes around important events; reward alone is insufficient evidence that “when” was learned.
- **Test phase-dependent behavior.** Check whether agents communicate differently during search, discovery, task execution, congestion/conflict, or after information becomes stale.
- **Add an actual reason to be selective.** Later experiments should impose a communication penalty/budget, latency, packet loss, range limit, or noise. Otherwise a cooperative environment may rationally learn to communicate all the time.
- **Scale team size and map size.** IC3Net's strongest results come from increasing scale; reproduce this style of evaluation in our environment rather than reporting one fixed configuration.
- **Role-specific gating is worth measuring.** For specialized roles, compare transmit rates and learned triggers by role. Different roles may rationally have different communication value even under a common team objective.
- **Do not treat individual rewards as automatically correct.** If we add per-agent shaping/auxiliary rewards, compare them against pure team reward and verify that they do not hard-code the desired communication behavior.
- **Compare against CommNet under identical optimization.** The baseline should use the same policy backbone, observation encoder, reward, rollout budget, and optimizer wherever possible so the gate is the actual changed factor.

7. Minimum comparison set implied by IC3Net

Variant	Question answered
No comm + global/team reward	How hard is the task without communication and with ambiguous credit?
No comm + individualized/auxiliary reward	How much does credit assignment alone help?
Always-on continuous comm + same reward	How much does WHAT-to-communicate help without timing decisions?
Gated continuous comm + same reward	Does learning WHEN to communicate improve performance/efficiency?
Gated comm + explicit cost/noise/delay	Does selective communication remain useful under realistic constraints?
Scale / role / observability variants	Does gating behavior generalize as the coordination problem changes?

8. Training details worth remembering, not blindly copying

- The paper uses 128-unit recurrent LSTMs, RMSProp, skip connections, and REINFORCE.
- Predator-prey and StarCraft are trained for 1000 epochs; traffic junction for 2000 epochs; results are averaged over five runs.
- Traffic junction uses curriculum learning to increase arrival difficulty over training.
- These settings are historical implementation details, not evidence that REINFORCE/RMSProp is optimal for a modern implementation.

Bottom line: IC3Net is valuable for the experimental idea that communication timing must be isolated and measured. The biggest lesson for us is not “use IC3Net exactly”; it is to disentangle communication gating from reward/credit assignment, include an always-on baseline, and only claim efficient selective communication when there is a genuine communication downside or budget.